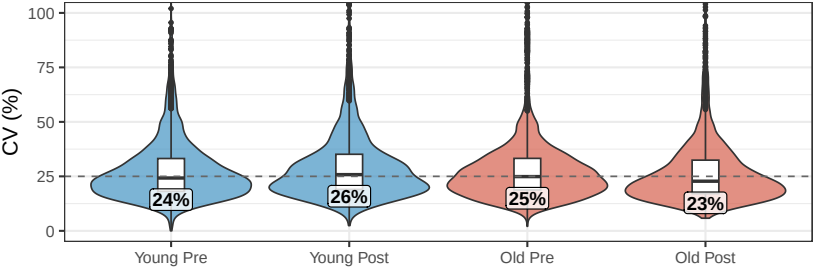


Proteomic Landscape of Aging and Resistance Training Response

Data quality, effect magnitude, contrast overlap, and functional enrichment across the 2 × 2 factorial design.

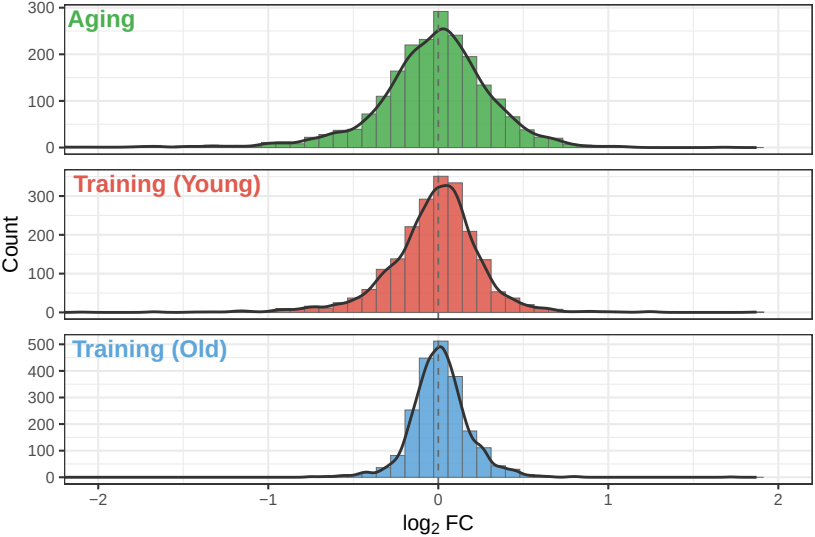
A Inter-Individual Variability (CV%)

Protein-level CV on cycloess-normalized intensities (linear scale)



B Effect Size Distribution (Log2FC)

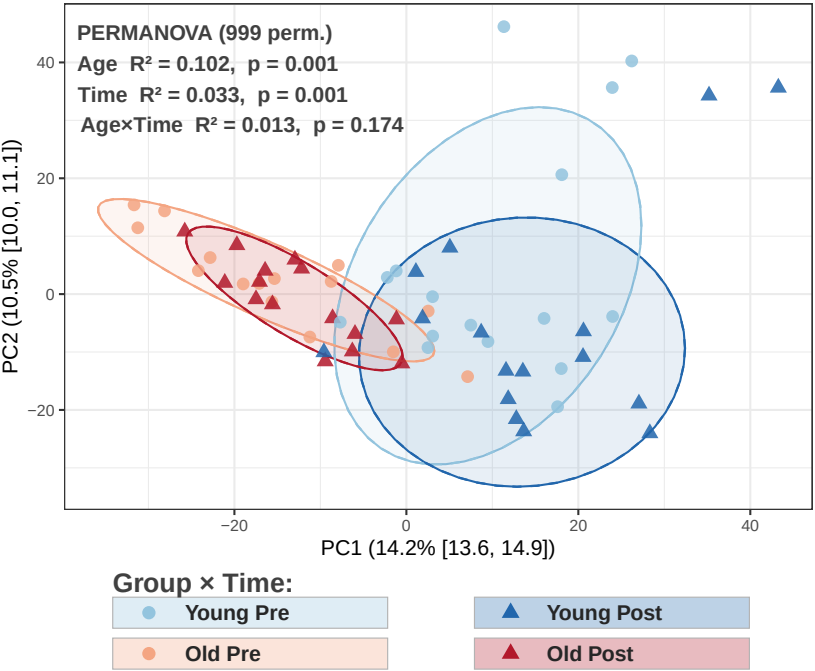
Med. |logFC|: Aging = 0.18 [0.17, 0.19], Tr. (Y) = 0.14 [0.13, 0.15], Tr. (O) = 0.09 [0.09, 0.10]
n(|logFC| > 0.5): Aging = 250, Tr. (Y) = 140, Tr. (O) = 29; all pairwise p < 0.001 (BH)



Young vs Old |logFC|: KS D = 0.18, p < 10⁻³⁰; Fligner χ^2 = 215, p < 10⁻⁴⁸

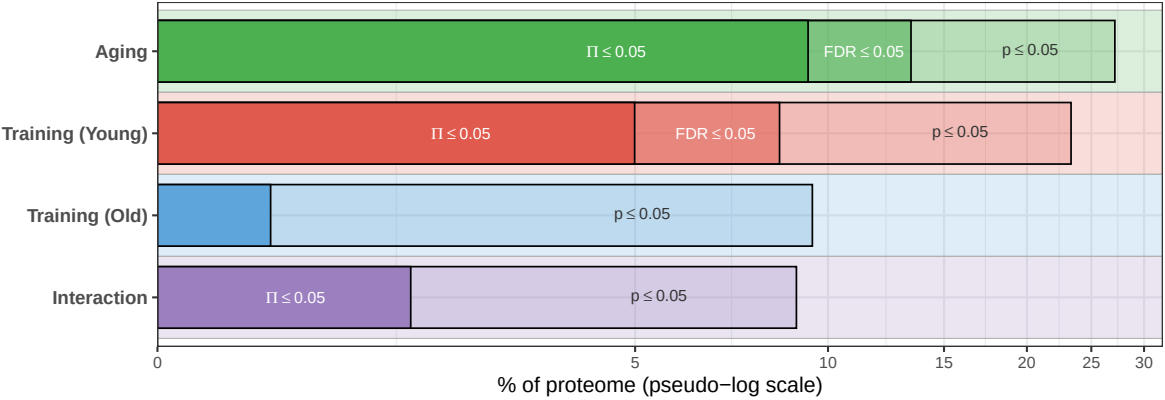
C Principal Component Analysis (PCA)

2,124 proteins (HPA-filtered, cycloess, BPCA-imputed); n = 62 samples; 80% confidence el

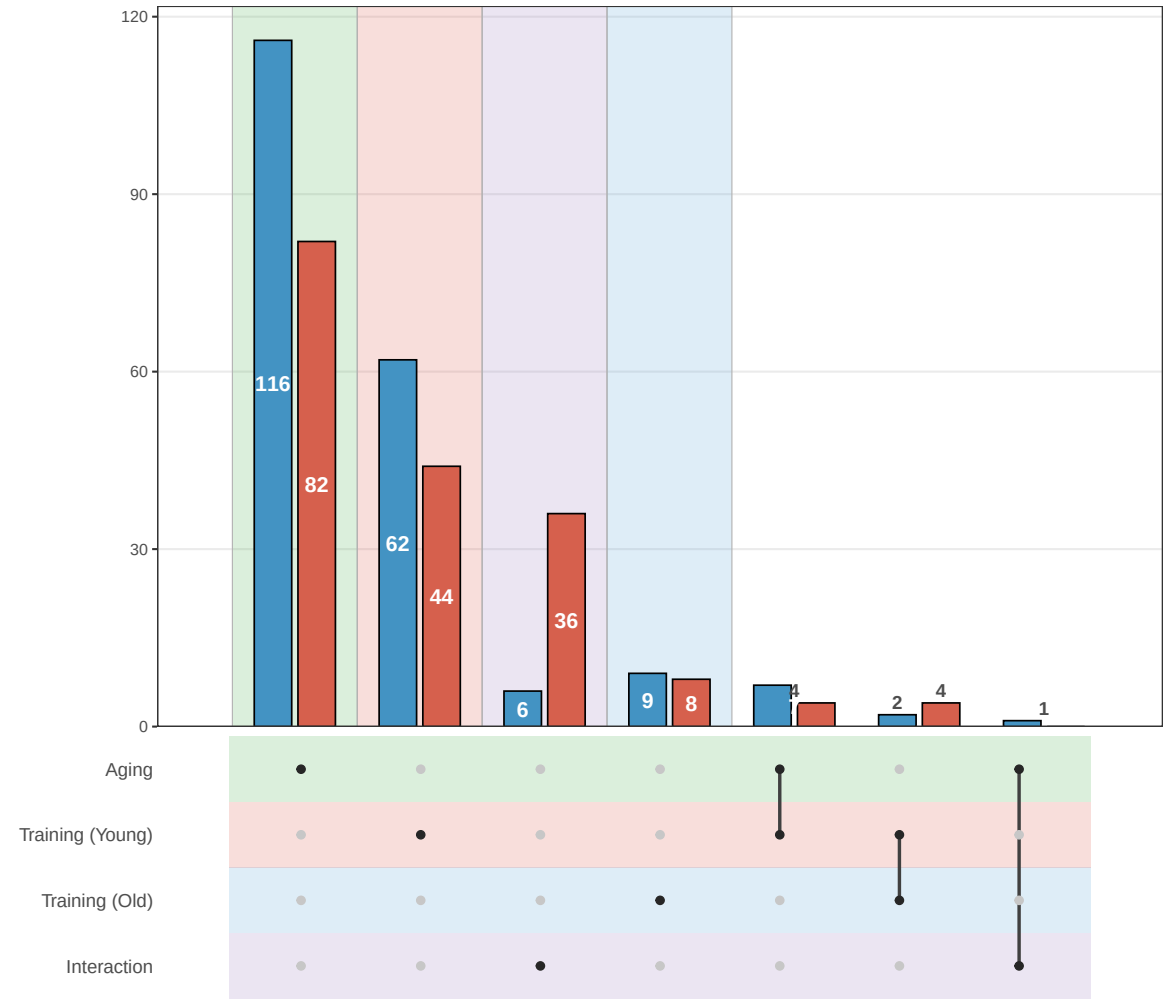


D DEPs per Contrast

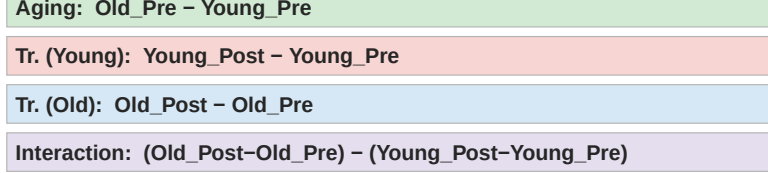
Fraction of 2,124 filtered proteins



DEPs by $\Pi < 0.05$; bars split by direction | Π : 363 / FDR: 464 total (59 mixed-direction excluded)



Contrast Definitions:



Direction:
Up
Down

F Pathway Enrichment

fGSEA (padj < 0.05); GO reduced via rrvgo (threshold 0.5)

